

Fulfillment Triage (SQL check)

Parameters

Time window (weeks)

26

▼

Minimum orders per category

200

- +

Top K categories

20

- +

Chatbot

Gemini model: gemini-3-flash-preview • Policy RAG: enabled (policy+ops_kb)

- Quick: Top late categories
- Quick: Pre-carrier or in-transit?
- Quick: Review risk checklist

 Hi! Tell me what you want to check (late delivery by category, stage clues, or review risk).

- Examples:
- Top late categories in the last 26 weeks
 - Is the delay mainly pre-carrier or in-transit?
 - What review risk signals should I check next?

No diagnosis plan yet. Ask a question above to generate one.

> Debug: Stage 4 status

Ask a question about fulfillment triage... (English)

↑

Late delivery structured interpretation output



Top late categories in the last 26 weeks



Got it — running **Late delivery by category** with current settings (weeks=26, min_orders=200, top_k=20). (router=llm)



Result preview (top 10):

	category_en	n_orders	n_late	late_rate	avg_days_late
0	health_beauty	3200	279	0.079	5.086
1	housewares	2149	152	0.0587	4.5132
2	bed_bath_table	2552	138	0.045	6.5217
3	watches_gifts	2174	130	0.0563	5.9154
4	furniture_decor	1646	125	0.0592	4.672
5	sports_leisure	1846	104	0.0507	5.8942
6	auto	1348	92	0.064	3.9239
7	computers_accessories	1667	91	0.049	5.2418
8	telephony	982	62	0.0578	4.8871
9	baby	909	55	0.0566	4.7091



Quick interpretation:

- **Summary:** Late delivery volume and rates are highest in the health and beauty category among the observed groups.
- **Signals (evidence):**
 - `health_beauty` has the highest volume of late orders at 279.0 and a `late_rate` of 0.079 [P2].
 - fields: `n_late`, `late_rate`
 - `bed_bath_table` exhibits the longest delay with 6.5217 `avg_days_late` from 138.0 `n_late` [P3].
 - fields: `avg_days_late`, `n_late`
 - `housewares` ranks high in late volume with 152.0 `n_late` across 2149 `n_orders` [P1].
 - fields: `n_late`, `n_orders`
 - `watches_gifts` shows 130.0 `n_late` and a `late_rate` of 0.0563 [P2].
 - fields: `n_late`, `late_rate`
- **Hypotheses (clues, not root cause):**
 - High late volume in certain categories may be caused by specific seller concentration risks [S2].
 - Categories with high average delays might be suffering from lane-specific transit issues [G1].
- **Checks (what to validate next):**
 - Identify if seller rankings in specific lanes are driving the volume of late orders [S2].
 - Execute a weekly trend analysis to identify if delays are persistent or due to seasonal spikes [S3]. Next action: Drill down lanes (seller_state → customer_state) for category='health_beauty' [P5]

Ask a question about fulfillment triage... (English)



Diagnosis plan for late delivery

Diagnosis plan

Executable 3-step workflow (domain-style).

1. Step 1 - Confirm impact

- Purpose: Confirm this is high-impact (volume $\times$ late volume), not only a high rate.
- Action: Validate `health_beauty` using `n_orders`, `n_late`, `late_rate`, `avg_days_late` in the preview table.

Mark checked

Marked as checked.

No table output for this step.

2. Step 2 - Localize by lane

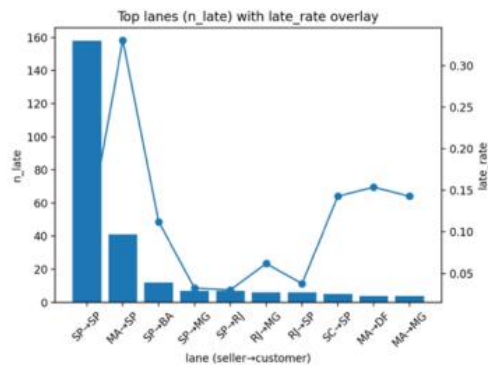
- Purpose: Check whether delays concentrate in a few seller+customer lanes (actionable routing).
- Action: Run lane drill-down (seller_state $\rightarrow$ customer_state).

Run step

Lane drill-down executed.

	seller_state	customer_state	n_orders	n_late	late_rate	avg_days_late
0	SP	SP	1092	158	0.0918	1.8101
1	MA	SP	103	41	0.3306	4.439
2	SP	BA	79	12	0.1121	8.5
3	SP	MG	172	7	0.0334	3.2857
4	SP	RJ	158	7	0.0302	16.7143
5	RJ	MG	68	6	0.0619	3.3333
6	RJ	SP	130	6	0.0377	2.1667
7	SC	SP	31	5	0.1429	5.6
8	MA	DF	23	4	0.1538	4.25
9	MA	MG	25	4	0.1429	4.5

Ops chart: Top lanes



3. Step 3 - Trend sanity check

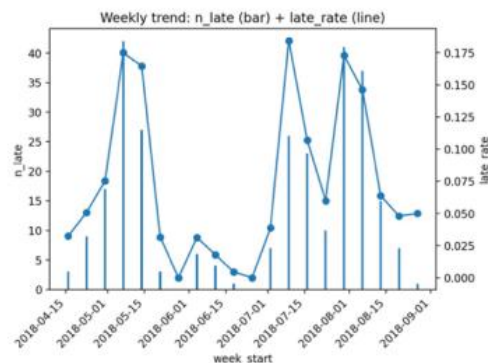
- Purpose: Confirm the signal is stable over time (avoid one-off spikes).
- Action: Run category weekly trend on late volume + late rate.

Run step

Weekly trend executed.

	week_start	n_orders	n_late	late_rate	avg_days_late
0	2018-04-16		87	3	0.0323
1	2018-04-23		164	9	0.0508
2	2018-04-30		195	17	0.0752
3	2018-05-07		214	42	0.175
4	2018-05-14		148	27	0.1646
5	2018-05-21		88	3	0.0316
6	2018-05-28		106	0	None
7	2018-06-04		180	6	0.0333
8	2018-06-11		195	4	0.0179
9	2018-06-18		203	1	0.0045

Ops chart: Weekly trend



Stage clue structured interpretation

Is the delay mainly pre-carrier or in-transit?

Got it — running Pre-carrier vs in-transit stage clues with current settings (weeks=26, min_orders=200, top_k=20). (router=llm)

Result preview (top 10):

	category_en	n_delivered	avg_h_purchase_to_approved	avg_h_approved_to_carrier	avg_h_carrier_to_customer
0	office_furniture	259	16.0917	272.9142	157.2249
1	home_construction	250	11.5171	118.4726	165.5616
2	pet_shop	638	9.1536	63.4074	137.8066
3	watches_gifts	2174	9.8719	61.7019	185.4253
4	furniture_decor	1646	12.918	58.7062	164.9393
5	computers_accessories	1667	10.7033	58.1885	170.7527
6	telephony	982	13.2731	57.6803	167.5648
7	cool_stuff	578	13.0687	54.7002	190.1541
8	health_beauty	3200	10.9436	53.4833	166.2051
9	baby	909	10.4356	53.4305	162.4943

Quick interpretation:

- **Summary:** The primary fulfillment bottleneck varies by category, with office_furniture showing pre-carrier friction while high-volume categories are primarily slowed by carrier transit times.
- **Signals (evidence):**
 - For office_furniture , the avg_h_approved_to_carrier reaches 272.9142, indicating significant pre-carrier friction compared to in-transit times [P7].
 - fields: avg_h_approved_to_carrier , avg_h_carrier_to_customer
 - High-volume categories like health_beauty with n_delivered of 3200 show in-transit times of 166.2051 dominating the fulfillment cycle [P8].
 - fields: avg_h_carrier_to_customer , n_delivered
 - Categories such as home_construction and office_furniture have lower n_delivered counts of 250 and 259, which might imply uncertainty in the stability of these averages [P3].
 - fields: n_delivered
- **Hypotheses (clues, not root cause):**
 - Extreme processing times in office_furniture may stem from warehouse handling complexities or stock replenishment delays [S1].
 - Consistent transit durations across multiple categories suggest that the primary bottleneck for most orders is likely within the carrier network [G1].
- **Checks (what to validate next):**
 - Examine seller performance metrics within office_furniture to isolate warehouse delays [S2].
 - Monitor weekly performance trends to identify if transit delays are increasing over time across all high-volume categories [S3]. Next action: In-transit hours by lane for category='office_furniture' [P5]

Ask a question about fulfillment triage... (English)



Diagnosis plan for stage clues

Diagnosis plan

Executable 3-step workflow (domain-style).

1. Step 1 — Identify dominant stage

- *Purpose:* Decide whether the symptom skews pre-carrier or in-transit (clue, not root cause).
- *Action:* Compare `office_furniture` using `avg_h_approved_to_carrier` vs `avg_h_carrier_to_customer` in the preview.

Mark checked

Marked as checked.

No table output for this step.

2. Step 2 — Localize by lane

- *Purpose:* Check whether the stage gap is concentrated in specific lanes.
- *Action:* Run lane drill-down (lane-level stage timings).

Run step

Lane drill-down executed.

	seller_state	customer_state	n_delivered	avg_h_purchase_to_approved	avg_h_approved_to_carrier	avg_h_carrier_to_customer
0	SP	RJ	39	11.7027	399.4775	
1	SP	SP	121	18.3966	268.5	

3. Step 3 — Trend sanity check

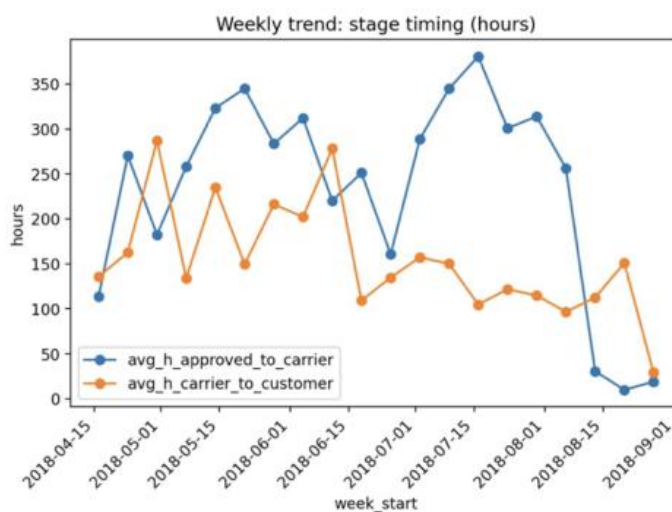
- *Purpose:* Check whether the stage gap persists under time slicing.
- *Action:* Run category weekly trend on stage timings.

Run step

Weekly trend executed.

	week_start	n_delivered	avg_h_purchase_to_approved	avg_h_approved_to_carrier	avg_h_carrier_to_customer
0	2018-04-16	13	45.5	113.4375	135.9
1	2018-04-23	19	23.0625	270.5	162.6
2	2018-04-30	15	8.8667	182.4667	21
3	2018-05-07	22	4.25	258.1875	134.
4	2018-05-14	20	12.6667	323.2917	235.0
5	2018-05-21	18	6.7308	345.0385	149.6
6	2018-05-28	12	1.7143	283.5714	216.2
7	2018-06-04	14	2.9412	312.2941	202.1
8	2018-06-11	10	15.0833	220.25	278.0
9	2018-06-18	18	20.8696	251.2609	109.3

Ops chart: Weekly trend



Ask a question about fulfillment triage... (English)



Review risk structured interpretation



What review risk signals should I check next?



Got it — running **Review risk by category** with current settings (weeks=26, min_orders=200, top_k=20). (router=llm)



Result preview (top 10):

	category_en	n_orders	n_low_ratings	low_rating_rate	avg_review_score	n_late	late_rate
0	bed_bath_table	2552	459	0.1496	4.0595	138	0.045
1	housewares	2149	346	0.1335	4.1604	152	0.0586
2	health_beauty	3200	324	0.0918	4.3366	279	0.079
3	watches_gifts	2174	320	0.1382	4.1154	130	0.0562
4	furniture_decor	1646	302	0.1431	4.1252	125	0.0592
5	computers_accessories	1667	239	0.1287	4.1773	91	0.049
6	sports_leisure	1846	196	0.0955	4.3345	105	0.0511
7	auto	1348	161	0.112	4.2584	92	0.064
8	telephony	982	147	0.1369	4.0899	62	0.0577
9	baby	909	107	0.1101	4.2482	55	0.0566



Quick interpretation:

- **Summary:** High order volumes in specific categories show a correlation between delivery delays and poor customer feedback across the fulfillment network.
- **Signals (evidence):**
 - The `bed_bath_table` category exhibits the highest volume of negative feedback with 459.0 `n_low_ratings` and a `low_rating_rate` of 0.1496 [P4].
 - fields: `n_low_ratings`, `low_rating_rate`
 - The `health_beauty` category has the largest volume at 3200 `n_orders` and the highest count of delayed shipments with 279.0 `n_late` [P3].
 - fields: `n_orders`, `n_late`
 - The `housewares` category shows a `late_rate` of 0.0586 and an `avg_review_score` of 4.1604, suggesting a potential correlation between delays and scores [P4].
 - fields: `late_rate`, `avg_review_score`
 - The `telephony` category has a `low_rating_rate` of 0.1369 with 982 `n_orders`, which is near the threshold for sample size considerations [P3].
 - fields: `low_rating_rate`, `n_orders`
- **Hypotheses (clues, not root cause):**
 - Volume heavy categories may experience fulfillment bottlenecks where delivery delays correlate with lower customer satisfaction scores [S1].
 - High negative feedback rates across the top categories suggest that shipping delays are a likely contributor to poor service quality [G1].
 - Concentration of delayed orders in specific segments suggests that localized lane issues may be driving the risk [S2].
- **Checks (what to validate next):**
 - Validate if specific shipping lanes are over represented in the late order counts for categories with high feedback risk [S2].
 - Review weekly trends to see if the deterioration in service quality is a stable or sudden occurrence [S3].
 - Compare processing times for the carrier and the customer delivery stages to identify the location of stage gap clues [G1]. Next action: Drill down lanes (`seller_state` → `customer_state`) for category=`'bed_bath_table'` [P5]

Ask a question about fulfillment triage... (English)



Diagnosis plan for review risk

Diagnosis plan

Executable 3-step workflow (domain-style).

1. Step 1 — Quantify risk exposure

- Purpose: Prioritize by impact (low-rating volume), not only by rate.

- Action: Validate `bed_bath_table` using `n_orders`, `n_low_ratings`, `low_rating_rate`, `avg_review_score` in the preview.

Mark checked

Marked as checked.

No table output for this step.

2. Step 2 — Localize by lane

- Purpose: Check whether low ratings cluster in specific lanes (possible service/fulfillment issue).

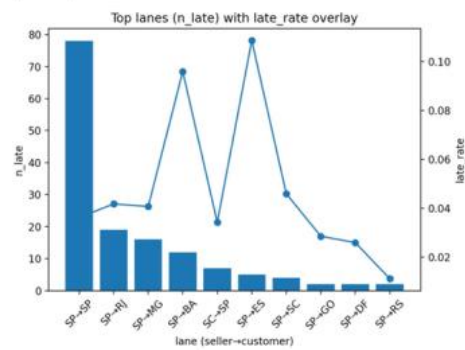
- Action: Run lane drill-down (seller_state → customer_state).

Run step

Lane drill-down executed.

	seller_state	customer_state	n_orders	n_late	late_rate	avg_days_late
0	SP	SP	1204	78	0.0363	4.4615
1	SP	RJ	271	19	0.0418	6.3158
2	SP	MG	221	16	0.0408	4.125
3	SP	BA	47	12	0.096	11.4167
4	SC	SP	101	7	0.0343	3.2857
5	SP	ES	38	5	0.1087	25.8
6	SP	SC	47	4	0.046	11.25
7	SP	GO	34	2	0.0286	3
8	SP	DF	38	2	0.026	4.5
9	SP	RS	109	2	0.0112	2.5

Ops chart: Top lanes



3. Step 3 — Cross-signal trend

- Purpose: Check if low ratings co-move with delivery performance over time (correlation ≠ causation).

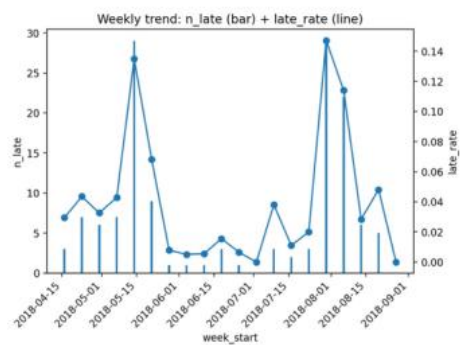
- Action: Run category weekly trend on low ratings + late rate.

Run step

Weekly trend executed.

	week_start	n_orders	n_late	late_rate	avg_days_late
0	2018-04-16	84	3	0.0294	11.3333
1	2018-04-23	134	7	0.0435	10.8571
2	2018-04-30	142	6	0.0326	8.5
3	2018-05-07	132	7	0.0429	18.1429
4	2018-05-14	171	29	0.1349	6.3793
5	2018-05-21	112	9	0.0682	6.1111
6	2018-05-28	111	1	0.0078	96
7	2018-06-04	160	1	0.0051	19
8	2018-06-11	157	1	0.0055	14
9	2018-06-18	170	3	0.0155	6.3333

Ops chart: Weekly trend



Ask a question about fulfillment triage... (English)

